11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

July 12, 2026

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. People who see our ads convert more — but the customers who *end up* seeing our ads are also the ones already leaning toward us (they browse, they search, they get retargeted). So **exposure is self-selected**, and a plain regression of sales on exposure is biased **upward**: it credits the ad for enthusiasm that was already there. We want the *true* causal effect of an extra exposure so we can set the right **budget** — the most we can pay per exposure and still come out ahead.

0.0.1 The problem in one word: endogeneity

When the treatment (ad exposure) is **correlated with the hidden drivers of the outcome** (a customer's unobserved intent), we say exposure is **endogenous**, and no amount of controlling for what we *measured* fixes it — the confounder (intent) is unmeasured. Notebook 05's adjustment tools are powerless here.

0.0.2 The fix: an instrument

An **instrumental variable (IV)** is a third variable Z that nudges the treatment *without* affecting the outcome any other way. Here Z is a randomized **encouragement** — e.g. a serving-priority lottery that makes some customers *more likely* to be shown the ad. Because Z is randomized, the slice of exposure it drives is as-good-as-random, uncontaminated by intent. IV isolates that clean slice and reads the effect off it. A valid instrument needs **four** things:

- **Relevance** — Z actually moves exposure (this one we *can* test: the **first-stage F-statistic**; F $<$ 10 is a "weak instrument" danger sign).
- **Exclusion** — Z affects sales *only through* exposure (the lottery itself sells nothing). Untestable from data — defended on design, and stress-tested in Step 6.
- **Exogeneity** — Z is independent of the hidden confounder (guaranteed here by randomization).
- **Monotonicity** — the encouragement never pushes anyone *away* from exposure (no "defiers": nobody is so annoyed by the nudge that they block ads in response). Untestable, but usually plausible for a gentle encouragement — and it is what buys the LATE interpretation below (derived in Step 3).

0.0.3 What IV recovers: a LATE

IV does **not** give the average effect over everyone. It gives the **LATE** (*Local Average Treatment Effect*) — the effect *for the "compliers,"* the customers whose exposure the encouragement actually changed. Always-exposed loyalists and never-exposed sceptics are silent in an IV estimate; report the scope accordingly.

> **On real data.** IV needs a genuine instrument, which is the hard part. Good marketing instruments: randomized **encouragement / intent-to-treat** designs (ship the nudge to a random half), ad-auction or delivery randomness, or a **cost shifter** for the price-elasticity problem in notebook 03. If you can randomize the *encouragement* (not the exposure itself), you have a clean instrument. Notebook **11b** (`11b_endogenous_exposure_real_data.ipynb`) runs this exact playbook on the real Criteo uplift data, where the randomized assignment instruments actual ad exposure — the real-world leg of this encouragement design.

0.1 2 · Simulate a ground truth

The **true** exposure->sales effect is **€15**. An unobserved `engagement` drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized `encouragement` (the instrument) nudges exposure without touching sales directly.

The data-generating model — exactly what `dgp.iv_ad_exposure` implements (defaults & seed in `src/cmp/dgp.py`). $n = 3000$; **unobserved** intent U = engagement $\sim \mathcal{N}(0,1)$; randomized encouragement $Z \sim$ Bernoulli$(\frac{1}{2})$; $\sigma(z) = 1/(1+e^{-z})$:

$$
\begin{aligned}
X = \text{exposure} &\sim \text{Bernoulli}(\sigma(0.3 + 1.1\,Z + 0.8\,U)) && \text{(first stage)}\\
Y = \text{sales} = 50 + 15\,X + 12\,U + \varepsilon, &\qquad \varepsilon \sim \mathcal{N}(0,\,6^2) && \text{(true effect: the 15).}
\end{aligned}
$$

U appears in **both** equations — $0.8\,U$ pushes exposure and $12\,U$ pushes sales — that is the self-selection, and since U is unobserved no regression adjustment can fix it (OLS is biased upward). All **four** IV conditions are visible by construction: **relevance** ($1.1 \neq 0$ — testable, the first-stage F below), **exclusion** (Z appears nowhere in the sales equation — untestable in real data, stress-tested in step 6), **exogeneity** (Z is a fair coin flip, drawn independently of U — randomization), and **monotonicity** ($1.1\,Z$ only ever *raises* the exposure probability — no defiers).

```
TRUE exposure->sales effect €15  ·  first-stage: exposure 56%->77% (shift +21
pp), F = 156
```

	encouragement	ad_exposure	sales
0	1.0	0.0	63.748992
1	1.0	1.0	64.636639
2	1.0	1.0	46.507627
3	0.0	1.0	75.599770
4	0.0	1.0	70.577754

0.2 3 · Identify — a valid instrument, and what IV recovers

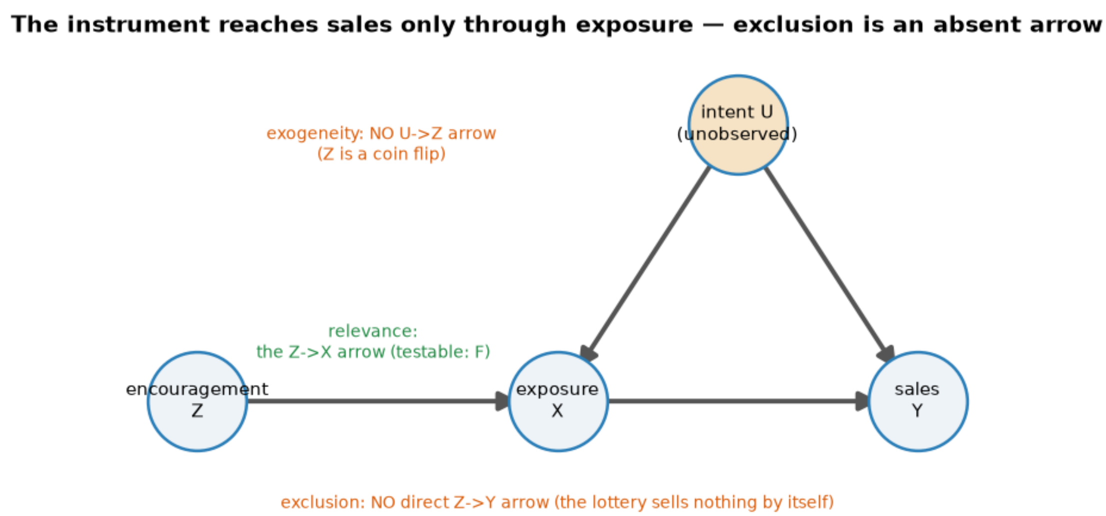
Exposure is **endogenous** — it is correlated with the outcome's hidden driver, Cov(exposure, $12U + \varepsilon) \neq 0$ (the U term specifically, since ε is independent structural noise) — so OLS is biased. The four requirements from the intro, now each named, formalized, and argued in marketing terms:

- **Relevance** — $\pi \neq 0$ in the first stage $X = b_0 + \pi Z + u$: the encouragement genuinely moves exposure. *Testable* — the first-stage F printed in Step 2 (F = 156 here; §5c defines F, explains the F < 10 rule, and shows what happens as an instrument decays).
- **Exclusion** — the encouragement has **no arrow of its own into sales**. In potential-outcomes notation, writing $Y_i(z, x)$ for customer i's sales if we *set* encouragement to z and exposure to x:

$$Y_i(z, x) \;=\; Y_i(x) \qquad \text{for all } z \in \{0, 1\} :$$

holding exposure fixed, flipping the lottery changes nothing. *Untestable from data* — defended on design (a serving-priority lottery sells nothing by itself; an encouragement *email that shows the product* would violate it), and stress-tested in Step 6. - **Exogeneity / independence** — $Z \perp (U, \varepsilon)$: the instrument is unrelated to the hidden drivers of exposure and sales. Holds here **by randomization** of the lottery. - **Monotonicity** — $X_i(1) \geq X_i(0)$ for every customer i, where $X_i(z)$ is i's exposure when the encouragement is set to z: the nudge never pushes anyone *away* from the ad (no defiers). *Untestable*; plausible for a gentle encouragement, and it is precisely what turns the IV ratio into an interpretable LATE — the derivation below uses it to kill one term.

The DAG states the structural part in one picture — and teaches that two of the assumptions are **missing arrows**:



Read the DAG by its absences. The unobserved intent U points into both exposure and sales — that fork is the endogeneity OLS trips over, and no arrow into U can be measured away. The instrument Z has exactly **one** outgoing arrow, into exposure: exclusion and exogeneity are not things Z *does*, they are arrows Z *does not have* — which is why they cannot be tested from data, only designed for. Relevance is the one arrow Z must have, and the only assumption the data can check.

Who is in the estimate? The four customer types. With $X_i(z) \in \{0,1\}$ the exposure customer i would end up with under encouragement z, two potential exposures give four types — the IV analogue of notebook 01's Persuadable / Sleeping-Dog grid:

	$X_i(1) = 1$ — exposed if encouraged	$X_i(1) = 0$ — unexposed if encouraged
$X_i(0) = 1$ — exposed anyway	**Always-taker** — brand loyalist who seeks the ad out regardless; the lottery changes nothing	**Defier** — so annoyed by the nudge they block the ad they'd otherwise have seen. *Ruled out by monotonicity*
$X_i(0) = 0$ — unexposed otherwise	**Complier** — the lottery tips them into seeing the ad. *The only type IV speaks for*	**Never-taker** — ad-blocker user, barely online; unreachable either way

The LATE theorem (Imbens–Angrist), in two lines. Decompose the **ITT** (intent-to-treat) effect on sales — the average sales difference between encouraged and not-encouraged customers, which *is* the reduced form — over the four types. Randomization of Z makes encouraged vs not comparable; **exclusion** says sales can only react to Z if exposure changed, which silences always- and never-takers ($X_i(1) = X_i(0)$); **monotonicity** deletes the defier term. What survives is the compliers:

$$\underbrace{E[Y \mid Z{=}1] - E[Y \mid Z{=}0]}_{\text{ITT on sales (reduced form)}} \;=\; E\big[(Y_i(1){-}Y_i(0))(X_i(1){-}X_i(0))\big] \;=\; \Pr(\text{complier})\cdot\underbrace{E[Y_i(1) - Y_i(0) \mid \text{complier}]}_{\text{LATE}}.$$

The same decomposition applied to exposure itself gives $E[X \mid Z{=}1] - E[X \mid Z{=}0] = \Pr(\text{complier})$ — **the first stage *is* the complier share** (the +21 pp printed in Step 2: about a fifth of these customers are compliers). Divide the two:

$$\text{LATE} \;=\; \frac{E[Y \mid Z{=}1] - E[Y \mid Z{=}0]}{E[X \mid Z{=}1] - E[X \mid Z{=}0]} \;=\; \frac{\text{reduced form}}{\text{first stage}} \;=\; \hat{\beta}_{\text{Wald}}.$$

So IV does **not** estimate the average effect over everyone: always-takers and never-takers cancel out of the numerator and are literally absent from the estimate — the LATE is the effect *for the customers the encouragement can move.* CausalPy fits the Bayesian version of this ratio: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity ρ rather than assuming it away.

> **Modeling note.** Because `ad_exposure` is binary (0/1), this joint MvNormal models the exposure first stage as a **linear-probability (Gaussian)** equation, so the two-equation model is an approximation — the exposure equation's posterior *width* shouldn't be read at face value; the reliable read-outs are its point estimate and the endogeneity ρ. §4b makes this misfit visible on purpose.

0.3 4 · Estimate — OLS, reduced form, and Bayesian IV

Three regressions tell the whole story, and seeing them side by side demystifies what IV actually does:

- **OLS** — regress sales on exposure directly. This is the **biased** number, inflated by self-selection.
- **Reduced form** — regress sales on the *instrument* (encouragement -> sales). This captures only the effect that flows through the clean, randomized channel.
- **First stage** — regress exposure on the instrument (encouragement -> exposure): how much the nudge moves exposure.
- **Wald / IV ratio** — reduced form ÷ first stage. The intuition: the instrument raised sales by *this* much and raised exposure by *that* much, so each unit of (instrument-driven) exposure is worth their ratio. That ratio *is* the causal effect. The **Bayesian IV** below is the same idea fit jointly with uncertainty (and it estimates the endogeneity ρ rather than assuming it away).

The fitted models, in symbols. The three frequentist point estimates:

$$\text{OLS: } Y = a_0 + a_1 X; \qquad \text{first stage: } X = b_0 + \pi Z; \qquad \text{reduced form: } Y = c_0 + \delta Z; \qquad \widehat{\beta}_{\text{Wald}} = \frac{\delta}{\pi};$$

and the Bayesian IV fits the first stage and the outcome equation **jointly**, with correlated errors:

$$\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} b_0 + \pi Z_i \\ \beta_0 + \beta X_i \end{pmatrix}, \, \Sigma \right), \qquad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\,\sigma_X \sigma_Y \\ \rho\,\sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix},$$

where the off-diagonal ρ **is** the endogeneity: the correlation between the two equations' errors that the hidden U induces and that OLS ignores. Priors: weakly-informative $\mathcal{N}(0, 50^2)$ on the regression coefficients (the code comment explains why the library default is too tight here) and an **LKJ prior** on Σ — a standard weakly-informative prior over correlation matrices that lets the data decide how strongly the two equations' errors move together (i.e. how big ρ is), rather than fixing it in advance.

Predict the OLS bias before running the regression. Because we built the DGM, we can compute exactly how wrong OLS will be — before it is wrong. With $Y = 50 + 15X + 12U + \varepsilon$, the OLS slope of Y on X converges to the planted effect **plus** whatever part of the error leaks in through the treatment (the **omitted-variable-bias** formula, specialized to this DGM):

$$\hat{a}_1 \xrightarrow{p} 15 + \frac{\operatorname{Cov}(X,\ 12U + \varepsilon)}{\operatorname{Var}(X)} = 15 + 12\,\frac{\operatorname{Cov}(X, U)}{\operatorname{Var}(X)},$$

since ε is independent noise. Even the sign is knowable in advance: the first stage loads on U with $+0.8$, so $\operatorname{Cov}(X, U) > 0$ and OLS must be biased **upward**. The code below re-draws the same simulation *retaining the normally-unobservable* U, evaluates this formula, and only then runs the regressions — theory should call the OLS number before the regression prints it. The same $\operatorname{Cov}(X, U)$ has a second life: it is what the Bayesian joint model estimates as a **positive error correlation** ρ between the exposure and sales equations.

```
OVB theory (computed before the regression):
  OLS ->p 15 + 12*Cov(X,U)/Var(X) = 15 + 8.5 = €23.5
OLS (biased)      €23.7  - the bias is computable self-selection, not bad luck
Reduced form (Z->Y) €3.5  ÷  first stage (Z->D) 0.21  =  Wald/IV ratio €16.5
Bayesian IV        €17.7  (true €15)  90% credible interval [€14.3, €21.1]
IV convergence: max r-hat 1.010 - min ESS 903 - divergences 0
Estimated endogeneity  rho(exposure err, sales err) = +0.22 [90% +0.10, +0.33]
  - nonzero rho is why OLS (€23.7) sits above IV (€17.7); it is the Bayesian
  image of the same Cov(X,U) the OVB formula priced at +€8.5.
```

Reading the sampler's two health checks. The Bayesian IV is fit by simulation — **MCMC**, the algorithm that draws samples from the posterior — so PyMC prints two numbers that say whether the sampler actually converged, and it will warn loudly if they look loose. **R-hat** compares the variance *within* each MCMC chain to the variance *across* chains; it should sit at ~1.00, and **<= 1.01 is the usual pass bar**. **ESS** (effective sample size) is roughly how many *independent* draws the autocorrelated chain is worth — a few hundred is ample for a posterior mean or interval. Under this notebook's FAST teaching profile the chains are deliberately short (only 200 draws x 2 chains to keep it quick), so R-hat can drift to ~1.04 and ESS can fall toward ~80, tripping PyMC's *"problems during sampling"* and *"effective sample size … smaller than 100"* notices. Those are **artifacts of the small draw count, not a broken estimate** — they clear in a FULL run, and they flag *how well we sampled the posterior*, never whether the IV logic is right. (The `90% credible interval` printed on the same line is the Bayesian analogue of a confidence interval: given the model and priors, there is a 90% posterior probability the true effect lies inside it — which is what lets us later say "the whole posterior sits above €10." The real backstop that the number is trustworthy is the multi-seed recovery-and-coverage check in §5b, not any single run's sampler line.)

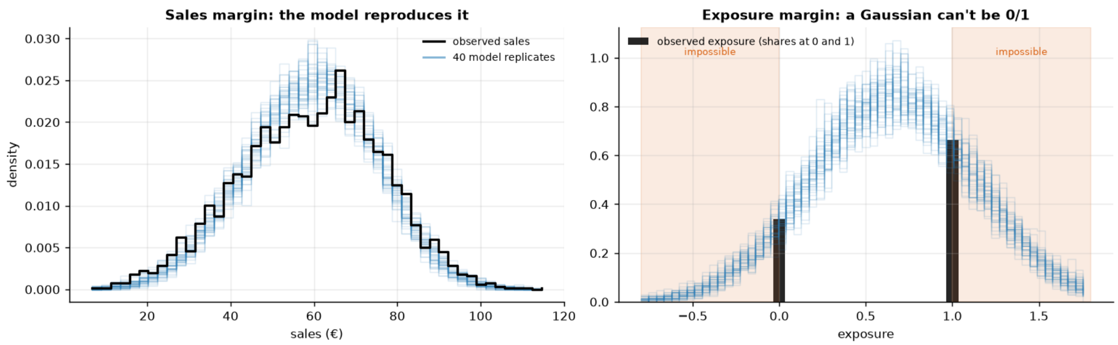
0.3.1 4b · Posterior predictive check — what the joint Gaussian gets right, and what it can't

Before leaning on the estimate we ask the fitted model to **reproduce the data it just saw**. For each of ~40 posterior draws we simulate a full replicate dataset from the fitted joint model,

$$(Y_i^{rep},\, X_i^{rep}) \;\sim\; \mathcal{N}_2\left(\begin{pmatrix}\hat\beta_0 + \hat\beta\, X_i \\ \hat b_0 + \hat\pi\, Z_i\end{pmatrix},\; \hat\Sigma\right),$$

(one $(\hat\beta_0, \hat\beta, \hat b_0, \hat\pi, \hat\Sigma)$ per draw) and overlay the replicated margins on the observed ones. The **sales margin should pass** — that is what licenses reading the causal β off the outcome equation. The **exposure margin cannot pass**: observed exposure is 0/1, and a Gaussian equation will cheerfully produce "exposures" of −0.3 or 1.4. We run it anyway, because that visible failure *is* the Step-3 modeling note, shown rather than footnoted.

```
Sales margin: replicate sd €15.7 vs observed €17.1, means €59.3 vs €59.3
  - close, with a mild under-dispersion (the joint approximation shows up
  even here, faintly): good enough to read beta off the outcome equation.
Exposure margin: 32% of replicated 'exposures' land outside [0,1]
  - impossible values, the linear-probability first stage made visible.
  Read that equation's point estimate and rho, never its width.
```



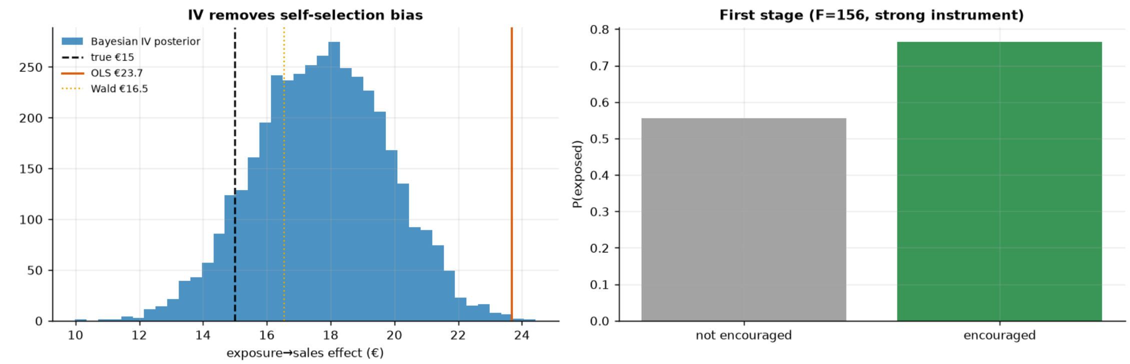
Read-out. *Left*: the replicate sales curves trace the observed distribution well — centred right, roughly the right width, if a hair narrower in the shoulders (the printed sd comparison quantifies it — even the "passing" margin carries a faint fingerprint of the approximation). That is a good enough fit to entitle us to read the causal β (and its interval) off the outcome equation. *Right*: the replicated "exposures" are smooth bell curves, and a substantial share of their mass sits in the shaded regions below 0 and above 1 — exposure values no customer can have (the printed line quantifies it). This is not a surprise to fix; it is the **linear-probability approximation made visible**, and it is exactly why §5b will find a small residual bias and mild under-coverage, and why the Step-3 modeling note told us to use the exposure equation's point estimate and ρ but never its posterior width. One diagnostic now carries what were two apologetic footnotes.

0.4 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both **remove that bias and land close to €15**, with the IV posterior's 90% interval **covering the truth**. The gap between OLS and IV **is** the self-selection bias the instrument removes. (We give the Bayesian IV weakly-informative priors so its interval reflects genuine sampling uncertainty — roughly the frequentist Wald width — rather than the artificially tight band CausalPy's default 2SLS-centred priors would produce.)

One honest note on LATE (and who the compliers are). IV recovers the effect for *compliers* — the customers whose exposure the encouragement actually moves — not the whole population. Under monotonicity that group is exactly the **first-stage shift: +21 pp (56% -> 77%)**, the estimated complier share. In *this* simulation every customer's true effect is €15 (the effect is homogeneous), so **LATE = ATE** and grading the IV against €15 is a fair test. On real data, where the effect varies across customers, IV would recover the complier-weighted LATE, which generally **!= the ATE** — and that is the right quantity for a budget call, since it answers *"what does an extra exposure do for the customers we can actually move?"*

```
self-selection bias removed: OLS €23.7 -> IV €17.7 (true €15)
```

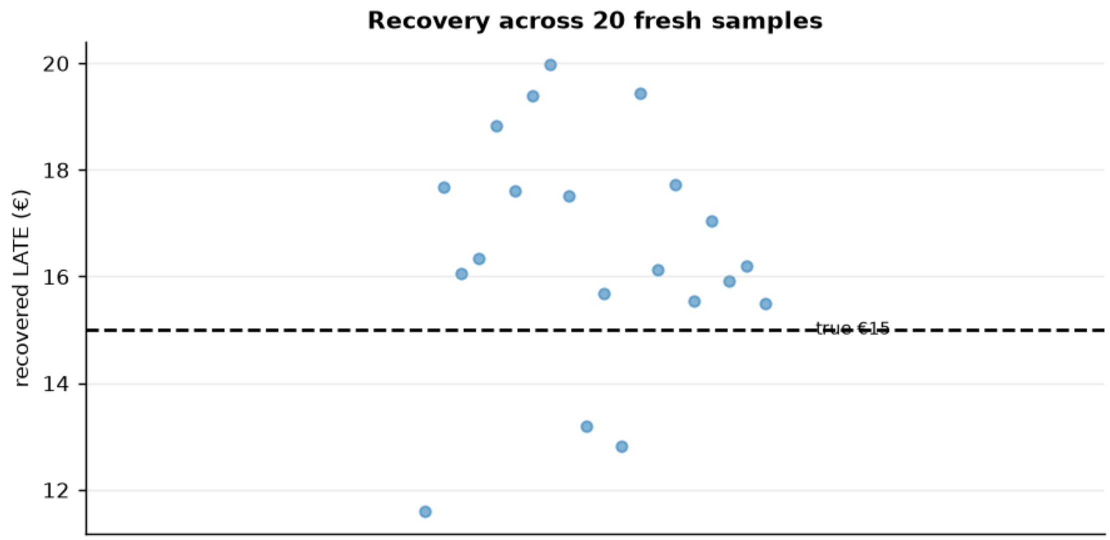


How to read this. *Left* — three vertical lines tell the whole story: **OLS (orange) sits well above the truth** (it credits the ad for pre-existing intent), while the **Bayesian IV posterior (blue) and the Wald ratio (gold) both drop back down toward the true €15**. The IV point carries a small upward finite-sample bias (a couple of euros above the truth on this seed; §5b measures the average across seeds), but its posterior — now honestly wide, thanks to weakly-informative priors — **covers €15** (just inside the lower edge); the artificially tight default-prior band would have excluded it. The horizontal distance from the orange line to the blue posterior *is* the self-selection bias — the euros of "effect" that were never causal. If you budgeted on OLS you'd over-spend on every exposure by exactly that gap. *Right* — the first stage: the encouragement clearly moves exposure (a big jump in P(exposed)), which is why the instrument is **strong** (F » 10). A weak first stage here would make the whole IV estimate unstable, so this panel is a precondition, not a footnote.

0.4.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each; their sampler chatter is silenced below, and we read only the aggregate bias/coverage tallies) and check both.

```
IV across 20 seeds: mean €16.5 (true €15)  bias +1.5  sd €2.1
  ·  90% interval covers truth in 16/20 seeds.
The committed seed (37) lands high (€18); across seeds the mean is €16.5
  - a small finite-sample bias of +1.5 € (~10%) toward OLS (the next cell asks
  whether it lives in the identification or in the model). The 90% interval
  covers €15 in 16/20 seeds (80% vs the 90% target - modest under-coverage,
  consistent with that bias); the budget verdict below never hinges on the
  last euro, which is why the ranking and the cap-style decision survive it.
```



Is the residual bias the identification's fault, or the model's? The Wald ratio *is* the IV identification with nothing else attached — two least-squares slopes and a division; no priors, no likelihood, no sampler. Re-running it on fresh draws at growing n therefore isolates what finite samples do to the identification itself. Whatever bias it does **not** show must belong to the Bayesian machinery around it — and §4b already fingered the suspect: the Gaussian first stage.

```
closed-form Wald across 30 seeds (identification only, no model):
  n =  1,000:  mean € 14.4  (bias -0.6)   spread (sd) €3.71
  n =  3,000:  mean € 15.1  (bias +0.1)   spread (sd) €2.53
  n = 10,000:  mean € 15.0  (bias +0.0)   spread (sd) €1.43
The Wald column stays pinned near €15 at every n - its deviation is within
sampling noise while the spread shrinks like 1/sqrt(n). The identification is
clean; the posterior-mean bias above therefore belongs to the Bayesian model's
finite-sample behaviour (the Gaussian first stage §4b exposed, plus
posterior-mean-of-a-skewed-posterior at modest draw counts), not to the IV
logic. §5c prices the pure-IV leakage via the 1/F rule at a few cents.
```

0.4.2 5c · What if the instrument were weak? (the headline pitfall, demonstrated)

Everything so far used a **strong** instrument: the encouragement moves exposure by ~21 pp and the first-stage F sits in the hundreds. The caveat "weak instruments are dangerous" deserves better than assertion, so we now degrade the instrument on purpose and watch the estimator come apart.

First, define the diagnostic we keep invoking. For a single instrument, the first-stage F is the ordinary F-statistic of the regression $X = b_0 + \pi Z + u$ (exactly what `est.first_stage_F` computes):

$$F = \frac{R^2/1}{(1-R^2)/(n-2)},$$

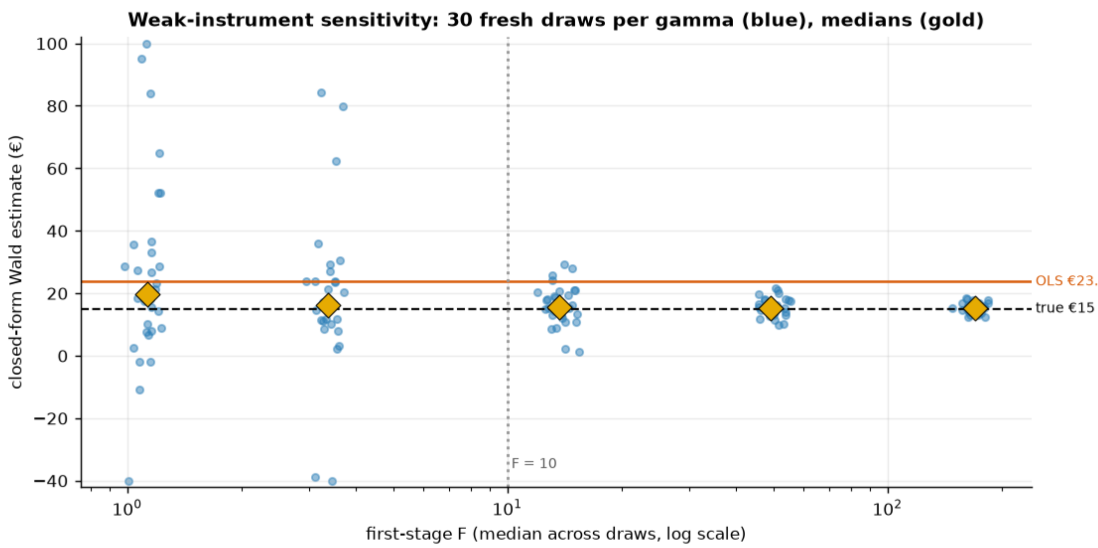
where R^2 is the share of exposure variance the instrument explains and $n-2$ counts the residual degrees of freedom: F measures the instrument's *signal* in the first stage relative to noise. **Why is the folklore threshold 10?** With one instrument, a standard approximation says the **2SLS** (two-stage least squares — the standard regression form of the same Wald ratio we computed above) / Wald estimator's residual bias, *relative to the OLS bias it is supposed to remove*, is roughly the reciprocal of F:

$$\frac{E[\hat{\beta}_{\text{2SLS}}] - \beta}{E[\hat{\beta}_{\text{OLS}}] - \beta} \approx \frac{1}{F},$$

so F = 10 means "no more than about a tenth of the OLS bias leaks back in" — a rule with teeth, not folklore. Cashed out on this notebook's numbers: F ~ 156 prices the pure-IV leakage at €8.7/156 ~ **€0.06** per exposure — negligible, which is why §5b's residual (more than an order of magnitude larger) had to come from the model approximation, not the instrument. (Caveat from `first_stage_F`'s own docstring: this is the classical **homoskedastic** F — it assumes the first-stage noise variance is constant across customers; on real, **heteroskedastic** data, where that variance changes across customers as real data usually does, prefer the Montiel Olea–Pflueger **effective F**, which corrects the F $<$ 10 threshold for exactly that non-constant noise, and treat F $>$ 10 as necessary, not sufficient.)

The experiment. `dgp.iv_ad_exposure` (src/cmp/dgp.py) hard-codes the encouragement coefficient at 1.1, so the cell below re-parameterizes the same six-line DGP with the first-stage strength γ as a dial — $X \sim \text{Bern}(\sigma(0.3 + \gamma Z + 0.8U))$, $Y = 50 + 15X + 12U + \varepsilon$ — and sweeps γ from this notebook's strength down to nearly-useless. For each γ we draw many fresh datasets and compute the **closed-form Wald only** (two `lstsq` calls per draw — the failure we are demonstrating is the identification's, so no MCMC is needed and the cell stays seconds-fast).

```
  gamma  1.1: median F  169.6  median Wald € 15.2  90% range [12.7, 18.0]
  gamma  0.6: median F   49.3  median Wald € 15.3  90% range [10.7, 20.6]
  gamma  0.3: median F   13.7  median Wald € 15.6  90% range [5.0, 27.0]
  gamma 0.15: median F    3.4  median Wald € 16.3  90% range [-20.4, 71.9]
  gamma 0.08: median F    1.1  median Wald € 19.8  90% range [-6.9, 90.2]
At F ~ 1.1 the 90% range of the IV estimate spans €-7 to €90 - far wider than
the €8.7 OLS-vs-truth gap it was meant to fix, and the median drifts toward OLS
(€23.7): a weak instrument is worse than none - shown, not asserted.
(3 extreme point(s) clipped to the plot window.)
```



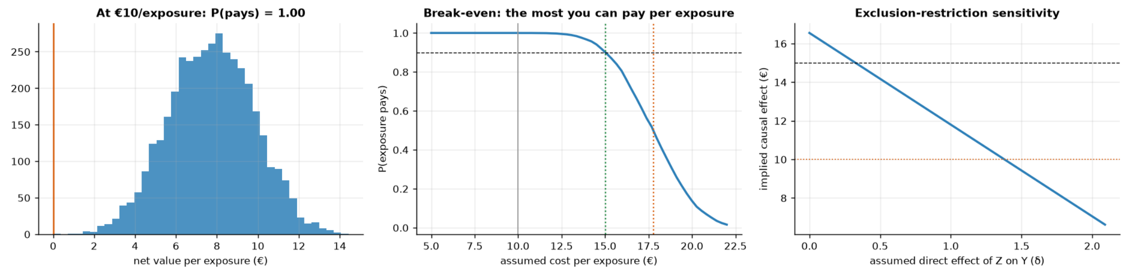
How to read this. Each blue column is the *same estimator on the same market* — only the instrument's strength changes. At the right edge (F in the hundreds, this notebook's regime) the Wald estimates hug the true €15. Moving left, the spread widens smoothly until, below the dotted F = 10 line, it explodes: single draws can land below zero or at multiples of the truth, and the gold *median* drifts up toward the orange OLS line. The mechanism is the 1/F formula read right-to-left — with a nearly-irrelevant instrument the Wald denominator is sampling noise around zero, so the ratio inherits OLS's bias *and* enormous variance. That is the precise sense in which **a weak instrument is worse than none**: OLS is wrong by a known-ish €9 with a tight interval, while a weak IV can be wrong by tens of euros in either direction *while wearing the costume of a causal estimate*. Lecture takeaway: check F before believing any IV number, and if F is small, walk away rather than report the ratio.

0.5 6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect δ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

```
At €10/exposure: causal net €7.7  ·  P(pays) 1.00
  -> BUY exposure (to the complier margin)
  P(pays)=1.00 is saturated, not assumed: the whole posterior
  [€14.3, €21.1] sits above €10.
Break-even sweep: exposure clears the 90%-confidence bar up to €15.0/exposure
  (the budget cap); at €17.8 it is a coin flip.
  So the €10 plan has ~€5.0 of headroom per exposure before the call gets tight.
Exclusion stress: the BUY call survives unless the instrument's direct effect
  exceeds delta~1.4 (out of a €3.5 reduced form) -
  trusting OLS (€24) would overstate the value and overspend.
```

```
(the corrected line is a point-estimate bound - it re-solves the Wald ratio
at posterior means; carry the IV interval's width alongside it when briefing
the cap.)
```



Read-out. Three numbers make the budget case, one per panel. *Left*: at the current €10/exposure the **causal** net value is ~ €8/exposure with the entire posterior above zero — a saturated BUY, valid for the complier margin the instrument identifies. *Middle*: the sweep turns the posterior into a procurement cap — the 90%-confidence bar holds up to ~ **€15/exposure**, the coin-flip point is ~ €17.8, so the €10 plan has ~€5 of genuine headroom. *Right*: the stress test bounds the untestable worry — the BUY survives unless the lottery email itself drives more than delta ~ €1.4 of sales directly, which against a €3.5 total reduced form would be a large leak (a plain encouragement email with no product content makes that implausible; an email that doubles as an ad does not — design the instrument accordingly). The management contrast to land: budgeting on OLS's €24 would sanction paying up to €24 for something causally worth ~ €15 — a €9 gap against the simulator's answer key. On real data the truth is unknown, so the wedge you'd actually avoid is OLS − IV ~ €6/exposure (the IV prices the exposure at ~ €17.7); either way, times every exposure bought, that is the price of mistaking correlation for incrementality — the cell below prices that same error at campaign scale (~ €5.9M).

0.5.1 The one-paragraph decision

> **What I'd tell the CMO.** Buy the exposure at the current €10: the causal value per exposure (the IV posterior above) sits comfortably clear of cost — P(pays) is saturated — and the break-even sweep caps procurement at ~ €15 per exposure at 90% confidence, coin-flip near €18, so the €10 plan carries roughly €5 of headroom. **Scope:** that number is earned on the *complier margin* — the ~ 21 pp of customers the encouragement actually tips into seeing the ad — and says nothing about always-exposed loyalists; that is the right scope for this call, because a budget buys *incremental* exposure. **Fragility, priced:** the BUY survives a direct-effect leak of the lottery into sales up to delta ~ €1.4 of the €3.5 reduced form (a large leak for a content-free lottery email), and survives the small finite-sample bias §5b measured; it would *not* survive a weak instrument — which is why F ~ 156 was checked before anything else (§5c shows the crash we avoided). **The mistake we didn't make:** budgeting on the naive €23.7 would sanction paying several euros of non-causal value on every exposure bought — the cell below prices that error at campaign scale, and the JSON carries every number a deck needs.

```
Campaign-scale cost of the naive number: an OLS-calibrated cap sanctions
  ~ €5.9 of non-causal value per exposure; at 1,000,000 exposures/quarter that
```

```
  authorizes ~ €5.9M per quarter for value that does not exist
  (€2.5M-€9.3M against the IV 90% interval).
```

```
{
  "true_effect": 15.0,
  "ols": 23.67,
  "wald": 16.55,
  "iv_mean": 17.73,
  "iv_ci90": [
    14.34,
    21.15
  ],
  "first_stage_F": 156.2,
  "complier_share_pp": 21.1,
  "rho_mean": 0.218,
  "cost_per_exposure": 10.0,
  "budget_cap_c90": 15.02,
  "coinflip_c50": 17.77,
  "exclusion_break_delta": 1.43,
  "multiseed_bias": 1.5,
  "multiseed_coverage": "16/20",
  "naive_overpay_eur_per_quarter_at_1M": 5940601
}
```


0.6 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves (the Step-3 table shows who is silent). Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first stage (low F) inflates both variance and bias — §5c demonstrated the crash: below F ~ 10 the estimate's spread explodes and its median drifts back toward OLS. Check F first (on real data, the Montiel Olea–Pflueger *effective* F), and if it is small, walk away rather than report the ratio.
- **Exclusion is untestable.** We formalized it ($Y_i(z, x) = Y_i(x)$), stress-tested it in step 6, and priced the leak that would flip the call; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating (it deleted the defier row in Step 3's derivation).
- **The joint Gaussian is an approximation.** §4b showed the exposure margin failing by construction (a 0/1 variable modeled linearly); §5b measured the resulting small bias and mild under-coverage. Read the exposure equation's point estimate and ρ, never its posterior width.
- **Real data next.** Notebook **11b** replays this playbook on the Criteo uplift dataset — randomized assignment as the instrument for actual ad exposure — where F, the complier share, and the budget cap must be earned from the data rather than planted in it.